

Analytical Problem Solving – Report

Artificial Intelligence (CSA17) • Assessment Tool 1

Overview

This report summarizes the approach, method, and result for each of the five problems in the Analytical Problem Solving assessment. Full working and explanations are provided in U1_A1_Solution.pdf; the algorithmic problems (Water Jug, 8-Queens, Logistics UCS) are additionally verified by the accompanying script U1_A1_Python_Code.py, whose console output is captured in U1_A1_Output.png.

1. Water Jug Problem

Method: Breadth-First Search over the space of (4-gallon, 3-gallon) jug states.

Metric	Value
Search strategy	Breadth-First Search
Moves to solution	6
Final state	4-gallon jug = 2, 3-gallon jug = 3

2. Mars Rover – Agent Analysis

Method: PEAS-style analysis of percepts, environment, actions, and performance measures, followed by architecture selection.

Property	Conclusion
Environment	Partially observable, dynamic, sequential, continuous, single-agent
Recommended architecture	Model-Based Utility-Based Agent

3. The 8-Queens Problem

Method: Backtracking search over column-by-column queen placement.

Metric	Value
Search strategy	Backtracking Search
Solution (Column→Row)	(1,4) (2,6) (3,8) (4,2) (5,7) (6,1) (7,3) (8,5)
Conflicts in final board	0

4. OLA Cab Booking – Agent Type

Method: Utility scoring of each cab category on comfort, fare, and wait time; highest-utility option is recommended.

Cab	Utility Score
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Micro	14.50
Mini	17.00
Sedan	20.00
Prime	24.50 (selected)
Shared	3.50

5. Logistics Network – Uniform Cost Search

Method: Uniform Cost Search over the weighted delivery graph.

Metric	Value
Start / Goal	S / G
Optimal path	S $\rightarrow$ A $\rightarrow$ D $\rightarrow$ G
Minimum cost	5

Conclusion

Each problem was solved using the search or agent-design strategy most suited to it – uninformed search (BFS, backtracking, UCS) for the well-defined search problems, and PEAS / utility-based reasoning for the two agent-design questions. The Python implementation independently reproduces the Water Jug, 8-Queens, and UCS results, confirming the correctness of the worked solutions.